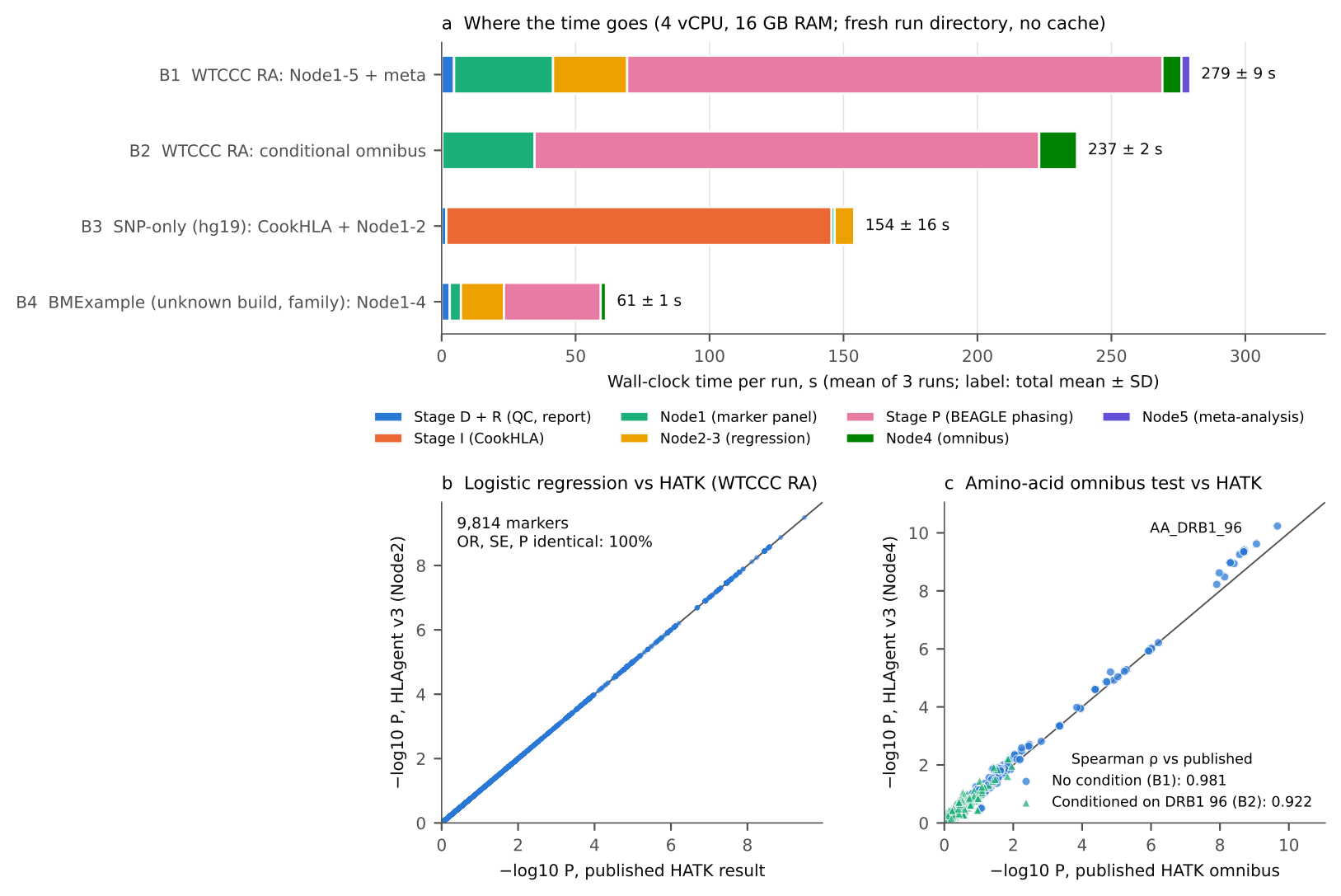

a Where the time goes (4 vCPU, 16 GB RAM; fresh run directory, no cache)
B1 WTCCC RA: Node1-5 + meta
B2 WTCCC RA: conditional omnibus
B3 SNP-only (hg19): CookHLA + Node1-2
B4 BMExample (unknown build, family): Node1-4
279 ± 9 s
237 ± 2 s
154 ± 16 s
61 ± 1 s
0
50
100
150
200
250
300
Wall-clock time per run, s (mean of 3 runs; label: total mean ± SD)
Stage D + R (QC, report)
Stage I (CookHLA)
Node1 (marker panel)
Node2-3 (regression)
Stage P (BEAGLE phasing)
Node4 (omnibus)
Node5 (meta-analysis)
b Logistic regression vs HATK (WTCCC RA)
9,814 markers
OR, SE, P identical: 100%
−log10 P, HLAgent v3 (Node2)
−log10 P, published HATK result
c Amino-acid omnibus test vs HATK
AA_DRB1_96
Spearman ρ vs published
No condition (B1): 0.981
Conditioned on DRB1 96 (B2): 0.922
−log10 P, HLAgent v3 (Node4)
−log10 P, published HATK omnibus